Analysis Of Capsule Network (Capsnet) Architectures And Applications

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Abstract—Image processing is a key to many automation problems. CapsNet are recent important contribution of Machine Learning field towards Image Processing task. In very short time they have surpassed results of old Neural Network methods and CNN their immediate predecessor. Here analysis of latest advances in CapsNet architecture and applications is studied. Also, latest advances in CNN are presented as it can inspire new architectures and applications of CapsNet. Results of latest architectures of both CapsNet and CNN are compared for various applications. Finally work concludes with future directions and potential new architecture and applications of CapsNet.

Keywords— CapsNet, CNN, Analysis, Applications of CapsNet, Architectures of CapsNet.

1. Introduction

Visuals are one of the most important inputs to humans. So, Machine Vision and Image Processing has become heart of automation. Machine Learning and Artificial Intelligence time to time contribute generously to this field. Now, in Machine Learning latest advancement of Capsule Networks (CapsNet) focus completely on machine vision and image processing. CapsNet are specialized neural networks (NN) to handle image features, and correlation between them [1]. It correlates at local as well as global level for better pattern matching and robustness. Also, it considers correlation between various parts of an image. CapsNet has successfully overpowered it immediate predecessor NN, Convolution Neural Networks (CNN). It has overcome drawbacks of CNN like insensitivity to feature position and rotation [1]. Here, in this work, latest advances in CapsNet since its inception are studied. Advances in architecture as well as application are covered here. Additionally, authors discuss latest advances of CNNs as this can inspire many new architectures and applications of CapsNet. Results of latest architectures of CapsNet and CNN are compared in the end.

2. Explaining Convolutional Neural Networks (Cnn)

2.1 LeNet

LeCun proposed this model for the first time as LeNet [2]. Then subsequently it was modified and used in various applications. Most popular and successful variations are AlexNet, and ResNet. Also, in this section latest CNN architectures are discussed.

2.2 AlexNet

AlexNet or ImageNet is an upgraded version of LeNet [3]. It had attached ReLU activation for each Convolution as well as fully connected layer. Also, this architecture is deeper than original one. It was designed in two pipelines so that it can be trained in parallel on multiple CPUs or GPUs. Stochastic Gradient Descent with momentum was used here for error back propagation. Overall, this architecture drastically reduced error in object detection and classification.

2.3 ResNet

Residual Neural Network Applied Convolution layers with skip paths [4]. Here architecture of Recurrent Neural Network is simulated. Architecture here consists of 152 layers and gave error rate less than human judge. As the skip path skips few layers so complexity of overall network is reduced in term of number of parameters processed.

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2.4 Latest Architectures

In [5] work, named AOGNet, grammar approach is combined with Convolutional networks. AND-operation represents concatenation of lower node outputs and OR-operation provides alternative combination paths. Additionally, terminal nodes are defined which take only sliced input. Proposed model is tested on benchmark datasets CIFAR-10, CIFAR-100 and ImageNet-1K. They have shown better performance than Residual Nets and Dense Net over classification and object detection tasks.

Work [6], proved that CNN can be trained to adapt to any domain of image by training on single image of that domain. Here once training is done using soriasis image data. Then other domain data is trained only using single image and comparable results are achieved.

Dilated Convolutional networks or atrous convolution Networks are proposed in [7]. Dilation refers to inserting holes in intermediate layer outputs to increase resolution. Better feature extraction is possible using this method. Squeeze net architecture is proposed by authors of [8]. Here intermediate layers use 1 by 1 convolution to reduce size of input. Then again 3 by 3 convolutions for expanding are done. By this process memory required and number of parameters required are reduced by 50 times. Results show squeeze net performs with same accuracy as Convolutional networks

Recurrent residual CNN with U-nets is proposed in [9]. It uses power of recur-rent, residual and U-nets neural networks together in CNN for image segmentation. Residual layer plays key role in training. Better feature recognition is possible using recurrent residual convolution layer. Additionally, this allows better U-net architecture which outperforms simple U-net and Residual U-nets. The model is tested on various datasets for segmentation task. Another work is [10] Images with poor ambient light are improved by using CNN. First input image is converted to a map and then ratio of image is used along with the map to dehaze the image using residual CNN.

Novel idea of Regional CNN is presented in [11]. Ultrasonic images of various sizes and orientations are successfully handled here for thyroid cancer detection. Here Regional Convolutional networks are improved by using multi-layer linking and multi-scale input. Experiments conclude with best number of iterations for high number of training samples, comparison of models based on AP, and ROC comparison. Proposed fast detection of thyroid cancer using region-based CNN has highest area under ROC curve compared to other CNN models. Global sum pooling (GSP) is novel method for pooling layer in CNNs [12]. Results here show that GSP performs better than traditional GLobal Average Pooling.

Evolving CNN is another architecture proposed by authors of [13]. Initially with small set of manually tagged dataset CNN is trained. Then by artificial methods images are collected from various sources. Using unsupervised ways, they are tagged. Later using these new images CNN is trained. If there is significant increase in performance beyond threshold then CNN weights are updated to accept new training. In all this process a standard validation image set is kept constant for verification benchmark. This framework of evolution is applied over four different CNN models. Over all models it has affected positively.

Another novel proposal is HSI-CNN for Hyper spectral Images from [14]. Authors here have developed an additional preprocessing layer before Convolutional neural network. This layer transforms hyper spectral image using one dimensional convolution. Then features obtained after one dimensional convolution are arranged in form of a two-dimensional matrix. This matrix is finally fed as input to Convolutional layer. This method has obtained promising results onto various datasets.

3. Capsule Networks (Capsnet)

Capsule Networks (CapsNet) were introduced recently for image processing tasks [1]. Sequence of multiple convolutional kernels bundled together is known as a capsule. Capsule Networks consists of large number of these capsules. Each capsule operates at local level of features. Global understanding of features is done by communication between various capsules using routing paths. Fig 1 shows architecture of Caps net as presented in [15]. This section presents what are the recent advances in CapsNet in terms of architectures and applications.

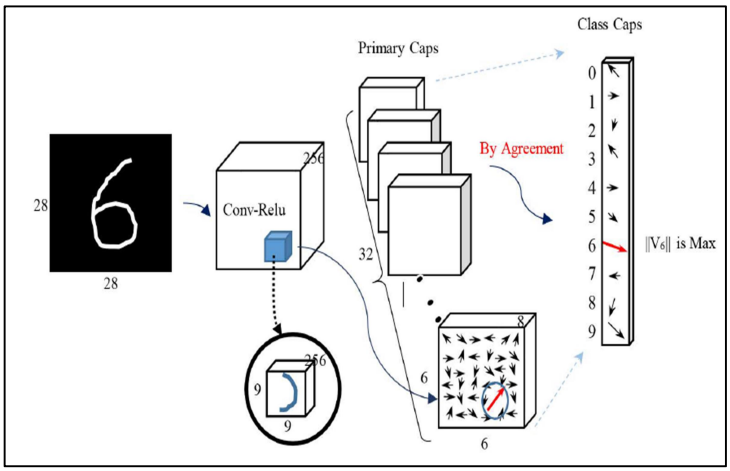

Figure 1. CapsNet Architecture Diagram

3.1 Latest Architectures

In Work [15], CapsNets are studied for better transparency of operations. It is argued that vectorized output of a capsule has likelihood information in regards to features. This enables ease in explain ability of capsule outputs. Also, authors propose how to improve transparency of CNNs using capsules in CNNs.

SegCaps, capsule network with Convolution-Deconvolution Capsules is presented in [16]. In this work, high demand for resources by CapsNets is handled. Proposed model uses routing within limited parent capsules as one way to restrict computations. On other hand it reduces size of shared matrices as well. All together affects drastically on memory and computational demand along with number of parameters being processed within layers. Deconvolution capsules use transformational convolution for keeping the global context for routes and parameters. Comparison shows how SegCaps operate on LUNA16 dataset with same effect with minimal parameters and resources for object segmentation.

Next architecture is Sparse Unsupervised Capsules from [17]. Supervised capsule networks have limitations in depth. Unsupervised capsule networks can be formed deeper layers. Here sparsely connected unsupervised capsules are used for classification problem. System is tested on AFFNIST dataset without training on it. This has given better benchmark results.

Another interesting advancement is brought out by [18]. Here unknown entities in text are handled efficiently. Bidirectional gated recurrent units are used before applying CapsNet. Here recurrent units find out possibility of relationship between entities. Then CapsNet uses this information to decide whether there exists hypernymy in compound entity or not. Fast CapsNet architecture is proposed in [19]. Here performance of CapsNet is improved using limited routing coefficients in each primary capsule. Single pixel in a primary capsule corresponds to single routing path. So, numbers of routing path are drastically reduced. Also, another important contribution of this work is convolutional decoder. This decoder helps in decoding samples with high visual variability in a restricted region.

Next, CapsuleGAN architecture [20] is novel step in generative adversarial networks (GAN). Here GAN objective function is also updated. Proposed model outperforms traditional convolution-based GAN in semi-supervised learning for MNIST and CIFAR10 datasets.

Another work [21] proposes CapsNet with Hit-or-Miss layer called as HitNet. Here new loss function called centripetal loss is proposed here. Hit or Miss capsules incorporate this loss function and they reduce losses when capsule is far from or too close to a class of sample supplied. Also, this architecture makes sure that gradient goes to zero only when class of capsule and sample is matched. Here authors claim that batch normalization process works better than method proposed for original CapsNet.

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Work [22] is another variation of CapsNet proposed recently. Here Long Short-Term Memory Neural Network is applied for evaluation. LSTM helps in predicting future position or transformation of the object based on history. Additionally, new routing is used to approximate an object's positions. Work has added another valuable part of evaluating transfer of learning for different datasets. Next is Recurrent Neural Networks (RNN) based Capsule network from [23]. Here capsules are formed based on RNN. This architecture allows interpreting time series data like language words. Also, it has proved efficient on sentiment analysis without prior knowledge of sentiment of words.

3.2 Latest Applications of CapsNet

Brain tumor type classification is considered as one of the toughest tasks in medical field. In [24] CapsNet are applied for this problem. This work uses CapsNets as they are thorough in handling various angles or scaling or other affine transformations on images. This work adapts CapsNet for brain tumor detection. Also, addresses overfitting problem occurring on real dataset. Here CNN and CapsNets are compared over complete brain and segmented tumor images. In both type of images CapsNet perform better than CNN. Another work [19] applies latest architecture of CapsNet for lung cancer detection.

Capsule Network Performance on Complex Data is studied in [25]. Authors here address problem of complex data from datasets like CIFAR10. Various models are formed here. Models of capsule networks are based on Ensemble layer, number of convolution layers and reconstruction scaling. Stacking of capsule layer is also applied as one model. All these models are compared with standard Convolution Neural Network and CpasNet. Here 7-model ensemble with additional convolutional layer is best with capsule network.

In thesis [26], detailed study of CapsNet is presented. Then it is applied for object comprehension in various views. Here 3D objects with various viewpoints are given for training. Then in training familiar and non-familiar viewpoints are provided. CapsNet show significant reduction in error rate compared to traditional CNN. On other hand [22] applied CapsNet for Object localization and motion transfer.

Sentiment Analysis is another area where CapsNet is applied [23]. CapsNet has proved to be very efficient in this application on three datasets. Here CapsNet is modified according to RNN architecture to support sentiment analysis application.

4. Results

Detailed analysis and comparison of CNN and CapsNet is presented in [27]. There it can easily be seen that CapsNet overpower CNNs for various parameters and various datasets. Also, it shows how Capsnet generalize over various datasets with different color, positions, transformations of objects and visibility. From literature surveyed, results and error rates on various datasets are taken. Results on MNIST dataset are taken from [19] and presented in fig 2. Here, it can be clearly seen that CapsNet outperforms the traditional CNNs. Also, Fastnet which is improvement to CapsNet does much better. On other hand error rate on various datasets is taken across literature and presented in fig 3. It proves that CapsNet can work well with various datasets.

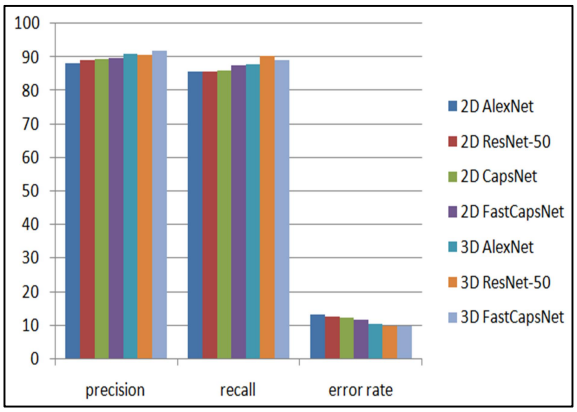

Figure 2. Comparison of various CapsNet architecture results on MNIST dataset

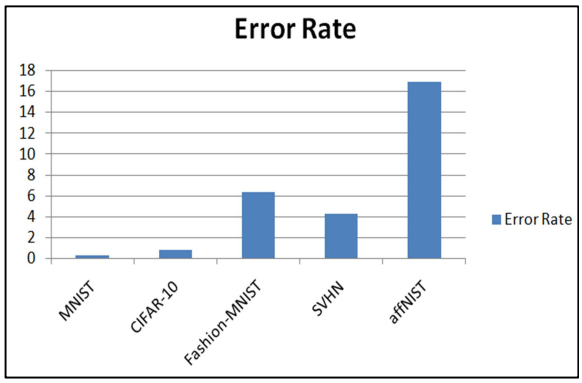

Figure 3. Error rate of CapsNet on various datasets

5. Conclusions And Future Scope

Recently proposed CapsNet has gained popularity in very short time. In nearly less than a year, it has been applied over various applications. Additionally, many modifications in architecture are proposed for it. In Image processing area researchers are working extensively on CapsNet and its predecessor CNN. CapsNet architecture uses Convolution as a basic component. So, applications in which CNN was successful recently or previously are always area for exploration in case of CapsNet. Further as CapsNet overcomes major drawbacks of CNN, such applications will lead to efficient systems. Recently CNN is applied on various areas. Similar areas and more can be extended for Capsule networks applications. So, based on this following can be some of good applications of CapsNets in near future.

Hyperspectral images from NASA are classified using Convolutional neural networks [14]. Problem handled here is multi-class classification problem. Value of a single output neuron is taken for training. Output was classified in total eight classes. Results of CNN were compared with SVM with and without sensitivity analysis. Both results show CNN performs better than SVM. Only for three minerals mica, calcite and graphite both techniques show nearly equal results. Image bias-based study on CapsNet can be carried out like in [28]. Real world image datasets used for face detection have biases which are not measured or controlled. This limits such datasets from calculating generalization ability of any face detection methods. Here in [28] synthetic datasets are generated with controlled biases to enable measurement of various parameters for face recognition systems. Here CNN's generalization ability is tested. CNN can generalize easily on any type of background but have limitation while generalizing pose of faces.

Various other areas where CapsNet can be applied are

– Rice classification [29]

– Lip reading [30]

–3D Human Action recognition using CNN with various body part and time-based features [31]

– 4D MRI Data Spatio-Temporal Data Classification [32]

– Object detection like real-time fire detection [33], and raindrop detection [34]

– Applications in Digital Pathology [35]

– DNA sequence Classification [36]

– ECG Classification [37]

References

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[1] S. Sabour, N. Frosst, and G. E. Hinton, "Dynamic routing between capsules," in *Advances in Neural Information Processing Systems*, pp. 3856–3866, 2017.

[2] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied todocument recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.

[3] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolu-tional neural networks," in *Advances in neural information processing systems*, pp. 1097–1105, 2012.

[4] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 770–778, 2016.

[5] X. Li, T. Wu, X. Song, and H. Krim, "Aognets: Deep and-or grammar networks for visual recognition," *arXiv preprint arXiv*:1711.05847, 2017.

[6] S. Valverde, M. Salem, M. Cabezas, D. Pareto, J. C. Vilanova, L. Rami´o-Torrent`a,` Rovira, J. Salvi, A. Oliver, and X. Llad´A.o, "One-shot domain adaptation in multi-ple sclerosis lesion segmentation using convolutional neural networks," *arXiv preprint arXiv:*1805.12415, 2018.

[7] J. Dolz, X. Xu, J. Rony, J. Yuan, Y. Liu, E. Granger, C. Desrosiers, X. Zhang, I. B.Ayed, and H. Lu, "Multi-region segmentation of bladder cancer structures in mri withprogressive dilated convolutional networks," *arXiv preprint arXiv:*1805.10720, 2018.

[8] F. N. Iandola, S. Han, M. W. Moskewicz, K. Ashraf, W. J. Dally, and K. Keutzer,"Squeezenet: Alexnet-level accuracy with 50x fewer parameters and¡ 0.5 mb model size,", *arXiv preprint arXiv:*1602.07360, 2016.

[9] M. Zahangir Alom, M. Hasan, C. Yakopcic, T. M. Taha, and V. K. Asari, "Recurrentresidual convolutional neural network based on u-net (r2u-net) for medical image segmen-tation," *arXiv preprint arXiv:*1802.06955, 2018.

[10] J. Li, G. Li, and H. Fan, "Image dehazing using residual-based deep cnn," *IEEE Access*,2018.

[11] Y. Wang, W. Ke, and P. Wan, "A method of ultrasonic image recognition for thyroidpapillary carcinoma based on deep convolution neural network," *NeuroQuantology*, vol. 16, no. 5, 2018.

[12] S. Aich and I. Stavness, "Object counting with small datasets of large images," *arXiv preprint arXiv:*1805.11123, 2018.

[13] Q. Xuan, H. Xiao, C. Fu, and Y. Liu, "Evolving convolutional neural network and itsapplication in fine-grained visual categorization," *IEEE Access*, 2018.

[14] F. Liang, H. Liu, X. Wang, and Y. Liu, "Hyperspectral image recognition based on artificialneural network," *NeuroQuantology*, vol. 16, no. 5, 2018.

[15] A. Shahroudnejad, A. Mohammadi, and K. N. Plataniotis, "Improved explainability ofcapsule networks: Relevance path by agreement," *arXiv preprint arXiv:*1802.10204, 2018.

[16] R. LaLonde and U. Bagci, "Capsules for object segmentation," *arXiv preprint arXiv:*1804.04241, 2018.

[17] D. Rawlinson, A. Ahmed, and G. Kowadlo, "Sparse unsupervised capsules generalize bet-ter," *arXiv preprint arXiv:*1804.06094, 2018.

[18] Q. Wang, T. Ruan, Y. Zhou, D. Gao, and P. He, "An attention-based bi-gru-capsnet modelfor hypernymy detection between compound entities," *arXiv preprint arXiv:*1805.04827,2018.

[19] A. Mobiny and H. Van Nguyen, "Fast capsnet for lung cancer screening," *arXiv preprint arXiv:*1806.07416, 2018.

[20] A. Jaiswal, W. AbdAlmageed, and P. Natarajan, "Capsulegan: Generative adversarialcapsule network," *arXiv preprint arXiv:*1802.06167, 2018.

[21] A. Deli`ege, A. Cioppa, and M. Van Droogenbroeck, "Hitnet: a neural network with cap-sules embedded in a hit-or-miss layer, extended with hybrid data augmentation and ghostcapsules," *arXiv preprint arXiv:*1806.06519, 2018.

[22] W. Liu, E. Barsoum, and J. D. Owens, "Object localization and motion transfer learningwith capsules," *arXiv preprint arXiv:*1805.07706, 2018.

[23] Y. Wang, A. Sun, J. Han, Y. Liu, and X. Zhu, "Sentiment analysis by capsules," in Proceedings of the 2018 World Wide Web Conference on World Wide Web, pp. 1165–1174, *International World Wide Web Conferences Steering Committee*, 2018.

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